Лабораторная работа №9

Настройка POP3/IMAP сервера

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Цель работы

Целью данной работы является приобретение практических навыков по установке и простейшему конфигурированию POP3/IMAP-сервера.

Установка Dovecot

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install dovecot telnet
Rocky Linux 9 - BaseOS                                                  15 kB/s | 4.3 kB     00:00
Rocky Linux 9 - AppStream                                              9.5 kB/s | 4.8 kB     00:00
Rocky Linux 9 - Extras                                                 7.1 kB/s | 3.1 kB     00:00
Dependencies resolved.
================================================================================================================
 Package                  Architecture        Version                          Repository                  Size
================================================================================================================
Installing:
 dovecot                  x86_64              1:2.3.16-15.el9                  appstream                  4.7 M
 telnet                   x86_64              1:0.17-85.el9                    appstream                   63 k
Installing dependencies:
 clucene-core             x86_64              2.3.3.4-42.20130812.e8e3d20git.el9  appstream              585 k
 libexttextcat            x86_64              3.4.5-11.el9                     appstream                  209 k

Transaction Summary
================================================================================================================
Install  4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm                          1.5 MB/s | 209 kB     00:00
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64.rpm     1.8 MB/s | 585 kB     00:00
(3/4): telnet-0.17-85.el9.x86_64.rpm                                  228 kB/s |  63 kB     00:00
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm                               2.4 MB/s | 4.7 MB     00:01
----------------------------------------------------------------------------------------------------------------
Total                                                                 2.2 MB/s | 5.6 MB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                        1/1
  Installing       : libexttextcat-3.4.5-11.el9.x86_64                                                      1/4
  Installing       : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64                                 2/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                                                         3/4
  Installing       : dovecot-1:2.3.16-15.el9.x86_64                                                         3/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                                                         3/4
  Installing       : telnet-1:0.17-85.el9.x86_64                                                            4/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64                                                         4/4
  Running scriptlet: telnet-1:0.17-85.el9.x86_64                                                            4/4
  Verifying        : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64                                 1/4
  Verifying        : dovecot-1:2.3.16-15.el9.x86_64                                                         2/4
  Verifying        : libexttextcat-3.4.5-11.el9.x86_64                                                      3/4
  Verifying        : telnet-1:0.17-85.el9.x86_64                                                            4/4

Installed:
  clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64   dovecot-1:2.3.16-15.el9.x86_64   libexttextcat-3.4.5-11.el9.x86_64
  telnet-1:0.17-85.el9.x86_64

Complete!
[root@server.user.net ~]#
```

Рис. 1. Открытие режима суперпользователя и установка пакета dovecot telnet.

Настройка dovecot

```
[user@server.user.net ~]$ postconf -e 'home_mailbox = Maildir/'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net ~]$
```

Рис. 2.1. Настройка в Postfix каталога для доставки почты.

Настройка dovecot

```
user@server:~

[user@server.user.net ~]$ mail_location = maildir:~/Maildir
bash: mail_location: command not found...
[user@server.user.net ~]$ mail_location = maildir:~/Maildir
bash: mail_location: command not found...
[user@server.user.net ~]$
[user@server.user.net ~]$ postconf -e 'home_mailbox = Maildir/'
postconf: fatal: open /etc/postfix/main.cf.tmp: Permission denied
[user@server.user.net ~]$ ^C
[user@server.user.net ~]$ sudo postconf -e 'home_mailbox = Maildir/'
[sudo] password for user:
[user@server.user.net ~]$ firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula bacula-cli
ent bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-
lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-uni
cast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client e
tcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangl
ia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc
 ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-cont
rol-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-services kube-schedule
r kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network ll
mnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql m
urmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole
ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dh
cp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-
master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-la
nsync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog s
yslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-
https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmp
p-local xmpp-server zabbix-agent zabbix-server zerotier
[user@server.user.net ~]$ firewall-cmd --add-service=pop3 --permanent
success
[user@server.user.net ~]$ firewall-cmd --add-service=pop3s --permanent
success
[user@server.user.net ~]$ firewall-cmd --add-service=imap --permanent
success
[user@server.user.net ~]$ firewall-cmd --add-service=imaps --permanent
success
[user@server.user.net ~]$ firewall-cmd --reload
success
[user@server.user.net ~]$ firewall-cmd --list-services
cockpit dhcpv6-client imap imaps ntp pop3 pop3s samba smtp ssh ssh-custom
[user@server.user.net ~]$
```

Рис. 2.2. Конфигурация межсетевого экрана, разрешив работать службам протоколов POP3 и IMAP.

Настройка dovecot

```
[root@server.claudely.net ~]#
[root@server.claudely.net ~]# restorecon -vR /etc
[root@server.claudely.net ~]# systemctl restart postfix
[root@server.claudely.net ~]# systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/sys
temd/system/dovecot.service.
[root@server.claudely.net ~]# systemctl start dovecot
[root@server.claudely.net ~]#
```

Рис. 2.3. Восстановление контекста безопасности в SELinux. Перезапуск Postfix и запуск Dovecot.

Проверка работы Dovecot

```
[root@server.claudely.net ~]#
[root@server.claudely.net ~]# MAIL=~/Maildir mail
s-nail: No mail for root at /root/Maildir
s-nail: /root/Maildir: No such entry, file or directory
[root@server.claudely.net ~]# doveadm mailbox list -u claudely
INBOX
[root@server.claudely.net ~]#
```

Рис. 3.1. Просмотр на терминале сервера имеющейся почты и mailbox пользователя.

Проверка работы Dovecot

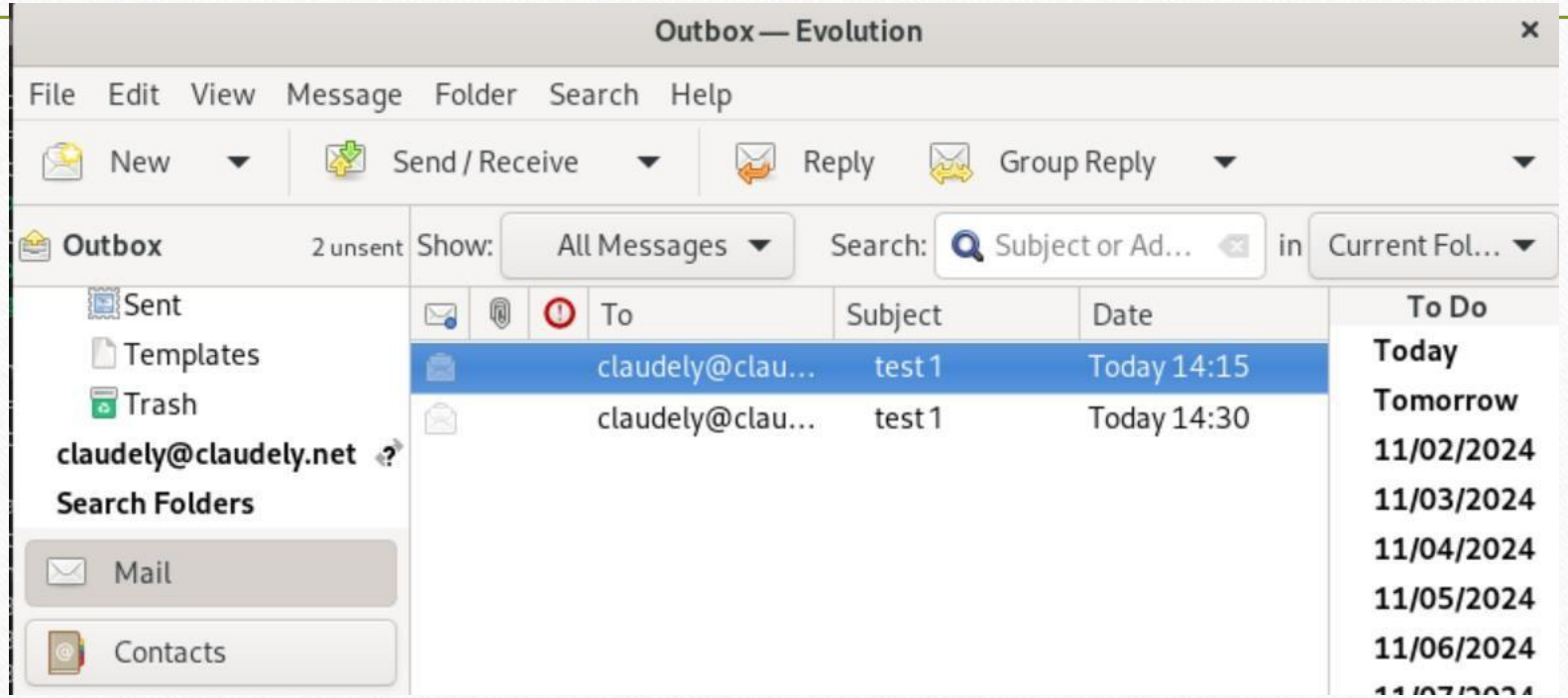

Рис. 3.2. Отправка из почтового клиента нескольких тестовых писем и проверка их доставки.

Проверка работы Dovecot

```
[root@server.claudely.net ~]# tail -f /var/log/maillog
Oct 31 13:16:06 server postfix/qmgr[48550]: 3CEC29252F9: from=<>, status=expired, returned
 to sender
Oct 31 13:16:06 server postfix/qmgr[48550]: 3CEC29252F9: removed
Oct 31 13:16:06 server postfix/smtp[49420]: 1F6CD925301: to=<root@client.claudely.net>, re
lay=none, delay=572916, delays=572916/0.13/0/0, dsn=4.4.3, status=deferred (Host or domain
 name not found. Name service error for name=client.claudely.net type=MX: Host not found,
try again)
Oct 31 13:16:06 server postfix/qmgr[48550]: 1F6CD925301: from=<>, status=expired, returned
 to sender
Oct 31 13:16:06 server postfix/qmgr[48550]: 1F6CD925301: removed
Oct 31 13:41:42 server postfix/postfix-script[50401]: stopping the Postfix mail system
Oct 31 13:41:42 server postfix/master[48173]: terminating on signal 15
Oct 31 13:41:42 server postfix/postfix-script[50477]: starting the Postfix mail system
Oct 31 13:41:42 server postfix/master[50479]: daemon started -- version 3.5.9, configurati
on /etc/postfix
Oct 31 13:42:07 server dovecot[50533]: master: Dovecot v2.3.16 (7e2e900c1a) starting up fo
r imap, pop3 (core dumps disabled)
```

Рис. 3.3. Просмотр сообщений, выдающихся при мониторинге почтовой службы на сервере.

Проверка работы Dovecot

```
  log          errors|find|reopen|test
  mailbox      cache|cache|create|delete|list|metadata|mutf7|path|rename|status|subscribe|
unsubscribe|update
  move         [-u <user>|-A] [-S <socket_path>] <destination> [user <source user>] <searc
h query>
  oldstats     dump|reset|top
  penalty      [-a <anvil socket path>] [<ip/bits>]
  process      status
  proxy        kick|list
  purge        [-u <user>|-A] [-S <socket_path>]
  pw           [-l] [-p plaintext] [-r rounds] [-s scheme] [-t hash] [-u user] [-V]
  rebuild      attachments
  reload
  replicator   add|dsync-status|remove|replicate|status
  save         [-u <user>|-A] [-S <socket_path>] [-m mailbox]
  search       [-u <user>|-A] [-S <socket_path>] <search query>
  service      status|stop
  sis          deduplicate|find
  stats        dump
  stop
  sync         [-u <user>|-A] [-S <socket_path>] [-1fPRU] [-l <secs>] [-r <rawlog path>] [
-m <mailbox>] [-g <mailbox_guid>] [-n <namespace> | -N] [-x <exclude>] [-s <state>] [-t <s
tart date>] -d|<dest>
  user         [-a <userdb socket path>] [-x <auth info>] [-f field] [-e <value>] [-u] <us
er mask> [...]
  who          [-a <anvil socket path>] [-1] [<user mask>] [<ip/bits>]
  zlibconnect  <host> [<port>]
[root@server.claudely.net ~]# mail
s-nail: No mail for root at /var/spool/mail/root
s-nail: /var/spool/mail/root: No such entry, file or directory
[root@server.claudely.net ~]#
```

Рис. 3.4. Просмотр сообщений, выдающихся при использовании doveadm и mail.

Проверка работы Dovecot

```
[root@client.claudely.net ~]#
[root@client.claudely.net ~]# postconf inet_protocols
inet_protocols = all
[root@client.claudely.net ~]# postconf -e 'inet_protocols = ipv4'
[root@client.claudely.net ~]# systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postf
ix.service.
[root@client.claudely.net ~]# systemctl start postfix
[root@client.claudely.net ~]#
```

Рис. 3.5. Подключение с помощью протокола Telnet к почтовому серверу по протоколу POP3 (через порт 110), ввод своего логина для подключения и пароля.

Внесение изменений в настройки внутреннего окружения виртуальной машины

```
[user@server.user.net server]$ cd /vagrant/provision/server
[user@server.user.net server]$ mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[user@server.user.net server]$ cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[user@server.user.net server]$ cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot
/conf.d
[user@server.user.net server]$ cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc
/dovecot/conf.d/
[user@server.user.net server]$ cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot
/conf.d/
[user@server.user.net server]$
```

Рис. 4.1. Переход на виртуальной машине server в каталог для внесения изменений в настройки внутреннего окружения /vagrant/provision/server/. Помещение в соответствующие подкаталоги конфигурационных файлов Dovecot и замена конфигурационного файла Postfix.

Внесение изменений в настройки внутреннего окружения виртуальной машины

root@server:/vagrant/provision/server root@server:~

```
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
dnf -y install dovecot telnet

echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc

echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --add-service=pop3 --permanent
firewall-cmd --add-service=pop3s --permanent
firewall-cmd --add-service=imap --permanent
firewall-cmd --add-service=imaps --permanent
firewall-cmd --reload

restorecon -vR /etc

echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
systemctl restart postfix
systemctl enable dovecot
systemctl start dovecot
```

Рис. 4.2. Внесение изменений в файл /vagrant/provision/server/mail.sh.

Внесение изменений в настройки внутреннего окружения виртуальной машины

```
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
dnf -y install evolution

echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'

echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
```

Рис. 4.3. Корректирование на виртуальной машине client в каталоге /vagrant/provision/client файла mail.sh.

Вывод

В ходе выполнения лабораторной работы были приобретены практические навыки по установке и простейшему конфигурированию POP3/IMAP-сервера.

Спасибо за внимание !